

Bridge test

Intro

On the opening day of the Millennium Bridge in London in the year 2000, hundreds of visitors crossing the bridge unintentionally induced lateral forces through their synchronized walking. This caused the bridge to begin wobbling sideways, a phenomenon that surprised engineers and shocked the public. As the oscillations increased, pedestrians adjusted their steps to maintain balance, which further amplified the movement. Due to safety concerns, the bridge was closed the following day to assess the situation and determine appropriate measures to ensure structural stability.

Test objective

The responsible engineers aimed to recreate the observed behavior under controlled conditions in order to precisely measure the bridge's natural frequency and understand the cause of the oscillations. The goal was to identify the mechanism behind the synchronization of pedestrian movement and its interaction with the bridge dynamics.

Test setup

On the day after the closure, engineers organized a controlled experiment. Friends, family members, and colleagues were invited to participate. Groups of people were systematically led across the bridge in increasing numbers while measurement equipment recorded vibrations and lateral displacements.

Results and conclusion

Following the closure, engineers began testing by gradually increasing the number of participants while monitoring the bridge's response. Up to 166 people, only slight and approximately linear oscillations were observed. However, beyond this threshold, the amplitude increased abruptly. The bridge underwent a phase transition into large lateral oscillations due to synchronized pedestrian movement. This successfully reproduced the original issue. As a solution, engineers installed damping systems to reduce lateral motion, and the bridge was later safely reopened.